National University of Computer and Emerging Sciences

SIGNAL AND SYSTEMS PROJECT REPORT

Title: Image Fusion using DWT

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1. Introduction:

Signal processing, image analysis and many more areas utilize the mathematical tool known as the Wavelet Transform. Unlike the Fourier Transform that split down a signal into sinusoidal waves, the Wavelet Transform divides a signal into components by both frequency and location, identifying precisely useful signals that feature localized details or abrupt changes.

One of the most known features of wavelet transformation is the ability to manage data at different levels of detail or several resolutions. This is accomplished by decomposing the signal, or image, into several frequency bands, each representing different scale details of the signal. Images are one complex signal with several smooth regions and sharp features and edges. The multi-resolution property offered by Wavelet Transform is perfect for image processing.

Wavelet transform has found a wide range of applications in many specialties. In image processing it is widely used for image compression, denoising, feature extraction, and image fusion. Towards medical imaging, images are enhanced using wavelets for better diagnosing accuracy. In geophysics and remote sensing, signals from radar and sonar systems are analyzed using wavelets. Their multi-resolution capability enables wavelets to extract details as and fine granularity.

2. Application: Image Fusion Using DWT:

Image fusion is a method that serves to merge several images of the same scene into one more informative and complete image. This application is very helpful when you have images from different sources or at different moments in time and wish to incorporate all the valuable information from all the images. The purpose is to strengthen essential features like details, clarity, and contrast that may be lacking in single images.

Here, in this project, we are employing the Discrete Wavelet Transform (DWT) technique in image fusion. DWT is a technique where an image is decomposed into various frequency bands, so that we can identify the significant details from the overall smooth regions of the image. This is the same way you would hear a song and pick up both the melody (which is constant and smooth) and the rhythm (which has sharp beats and rapid changes).

Here's the process of how DWT works in image fusion:

- Decomposition of Images: The DWT breaks each image into two broad categories of components:
- High-frequency components: These have the finer details of the image, such as edges, textures, and fine features. For instance, in a landscape picture, the distinct edges of trees, mountains, and buildings are high-frequency details.
- Low-frequency components: These constitute the smooth general structure of the image, such as the sky, background, or the large uniform regions in the image.
- Combination of Elements: After decomposing the images, we apply various methods to merge the high-frequency and low-frequency elements. For example:
- MaxMax Fusion: For high-frequency details, we take the maximum value from both images so that the most salient features are retained.
- MinMax Fusion: On low-frequency terms, we could utilize the maximum or minimum terms from the pair of images in order to get to keep major structural details.

By fusing the high-frequency details and low-frequency structures of both images, we can create a fused image that includes all the significant information from both sources. This image will be more detailed and have clearer structural features, hence being more informative than the two individual images.

This method is particularly beneficial in disciplines such as medical imaging, in which physicians may employ images obtained from various modalities (i.e., MRI and CT scans) to acquire a better impression of a patient's status. With image fusion utilizing DWT, significant characteristics from each of the images are merged to obtain a more complete picture.

3. Methodology:

In this project, we used various techniques like image fusion, image morphing, image mixing, and image restoration using Wavelet Transforms. These techniques were employed to process images in varied manners. Here is a step-by-step explanation of the methodology for each of them:

Image Fusion Using DWT

Image fusion refers to the integration of two or more images into one more informative image. For our project, we utilized Discrete Wavelet Transform (DWT to decompose and merge significant details of images:

- Image Decomposition: The image is decomposed initially into various frequency bands by applying DWT. This decomposition decomposes the image into:

Low-frequency components (which retain the overall structure of the image, such as backgrounds and large areas). High-frequency components (which retain the small details of the image, such as edges and textures).

- Fusion of Components: Once the images are decomposed, we employ a fusion method to merge the low and high-frequency components of both images. Various fusion strategies such as MaxMax or MinMax are used; MaxMax: In the case of high-frequency details, we choose the max values from both the images to maintain the sharpest details. MinMax: In the case of low-frequency components, we choose the minimum or maximum values to keep the most significant overall structures.
- Image Morphing: Morphing an image is a process of transforming an image into another in a smooth manner. We used this process in our project to merge two images into each other so that the images changed over from one to another seamlessly. This is achieved by a gradual change of pixels from one image to the other to form a smooth animation or change.
- Image Mixing: Image blending refers to the procedure of fusing two or more images to obtain a new image with combined attributes. In our project, we used a simple method to blend images with varied weightages to each image to enable us to control the proportion of contribution of each source image.
- Image Restoration: Image restoration refers to enhancing or retrieving degraded images because of noise, blurring, or other types of distortion. We restored the details and clarity of an image by removing unnecessary noise and smoothening out the distortions in our project using DWT-based techniques. We can do this by isolating the high-frequency components corresponding to the aspects concerned and nullifying the noise to get a better image.

All these techniques enabled us to handle images in various ways, enabling enhanced visualization and analysis of the images for various uses such as medical imaging, surveillance, or photography.

4. Detailed Design and Architecture

The system architecture is built around the concept of wavelet decomposition and fusion. The key components of the system are:

- Image Input: Two images are taken as input from the user.
- Wavelet Decomposition: The images are decomposed into sub-bands using the wavelet transform (using MATLAB's `dwt2()` function).
- Fusion Logic: Fusion strategies such as MeanMax, MinMax, and others are applied to the sub-bands.

- Inverse Wavelet Transform: The inverse DWT (`idwt2()`) is applied to reconstruct the fused image from the fused sub-bands.

The system ensures that only the relevant details from both images are preserved and merged to create an enhanced image. The process is demonstrated with both images being shown before and after fusion.

5. Implementation, Testing, and Programming Code:

```
%Program for Image Fusion
clear;
fusiontype='MaxMax';
wavetype='coif5';
[filename1, pathname1] = uigetfile({'*.jpg;*.png;*.bmp','Image Files (*.jpg, *.png, *.bmp)'}, 'Select the 1st Image');
if isequal(filename1,0)
    error('No file selected for Image 1.');
end
filewithpath1 = fullfile(pathname1, filename1);
img1 = imread(filewithpath1);
filewithpath1=strcat(pathname1,filename1);
img1=imread(filewithpath1);
%Loading image 2.
[filename2, pathname2] = uigetfile({'*.jpg;*.png;*.bmp','Image Files (*.jpg, *.png, *.bmp)'}, 'Select the 2nd Image');
if isequal(filename2,0)
    error('No file selected for Image 2.');
end
filewithpath2 = fullfile(pathname2, filename2);
img2 = imread(filewithpath2);
filewithpath2=strcat(pathname2, filename2);
img2= imread(filewithpath2);
[row, col]=size(img1(:,:,1));
if ~isequal(size(img1),size(img2))
    img2=imresize (img2, [row,col]);
end
fusedimageR=imgfusion(img1(:,:,1),img2(:,:,1),fusiontype, wavetype);
fusedimageG=imgfusion(img1(:,:,2),img2(:,:,2), fusiontype, wavetype);
fusedimageB=imgfusion(img1(:,:,3),img2(:,:,3), fusiontype, wavetype);
fusedimage=uint8(cat (3, fusedimageR, fusedimageG, fusedimageB));

imwrite(imresize(fusedimage, [row,col]), fullfile(tempdir, 'fusedimage.jpg'), 'Quality', 100);
disp(['Image saved at ', fullfile(tempdir, 'Cathefused.jpg')]);

subplot(131)
imshow(img1)
title('Image1')
subplot(132)
imshow(img2)
```

```
title('Image2')
subplot(133)
imshow(fusedimage)
title('Fused Image')
%Function to create fused image
function outimage=imgfusion(Im1, Im2, ftype, wtype)
[cAl,cH1,cV1,cD1]=dwt2(double(Im1),wtype, 'per');
[cA2,cH2,cV2, cD2]=dwt2 (double(Im2), wtype, 'per');
[r,c]=size(cAl);
cA=zeros(r,c);
cH=zeros(r,c);
cV=zeros(r,c);
cD=zeros(r,c);
switch ftype
    case 'MeanMean'
        for i=1:r
            for k=1:c
                cA(i,k)=mean ([cAl(i,k),cA2(i,k)]);
                cH(i,k)=mean([cH1(i,k),cH2(i,k)]);
                cV(i,k)=mean([cV1(i,k),cV2(i,k)]);
                cD(i,k)=mean([cDl(i,k),cD2(i,k)]);
            end
        end
    case 'MeanMax'
            for i=1:r
                for k=1:c
                    cA(i,k)=mean ([cA1(i,k),cA2(i,k)]);
                    cH(i,k)=max([cH1(i,k),cH2(i,k)]);
                    cV(i,k)=max([cV1(i,k),cV2(i,k)]);
                    cD(i,k)=max([cD1(i,k),cD2(i,k)]);
                end
            end
    case 'MeanMin'
        for i=1:r
            for k=1:c
                cA(i,k)=mean([cAl(i,k),cA2(i,k)]);
                cH(i,k)=min([cH1(i,k),cH2(i,k)]);
                cV(i,k)=min([cV1(i,k),cV2(i,k)]);
                cD(i,k)=min([cD1(i,k),cD2(i,k)]);
            end
        end
    case 'MaxMin'
        for i=1:r
            for k=1:c
                cA(i,k)=max([cAl(i,k), cA2(i,k)]);
                cH(i,k)=min([cH1(i,k),cH2(i,k)]);
                cV(i,k)=min([cV1(i,k),cV2(i,k)]);
                cD(i,k)=min([cD1(i,k),cD2(i,k)]);
            end
        end
    case 'MinMean'
        for i=1:r
            for k=1:c
                cA(i,k)=min([cAl(i,k),cA2(i,k)]);
                cH(i,k)=mean([cH1(i,k),cH2(i,k)]);
```

```
                cV(i,k)=mean([cV1(i,k),cV2(i,k)]);
                cD(i,k)=mean([cD1(i,k),cD2(i,k)]);
            end
        end
    case 'MinMax'
        for i=1:r
            for k=1:c
                cA(i,k)=min([cAl(i,k),cA2(i,k)]);
                cH(i,k)=max([cH1(i,k),cH2(i,k)]);
                cV(i,k)=max([cV1(i,k),cV2(i,k)]);
                cD(i,k)=max([cD1(i,k),cD2(i,k)]);
            end
        end
    case 'MinMin'
        for i=1:r
            for k=1:c
                cA(i,k)=min([cA1(i,k),cA2(i,k)]);
                cH(i,k)=min([cH1(i,k),cH2(i,k)]);
                cV(i,k)=min([cV1(i,k),cV2(i,k)]);
                cD(i,k)=min([cD1(i,k),cD2(i,k)]);
            end
        end
end
outimage = uint8((double(Im1) + double(Im2)) / 2);

end
```

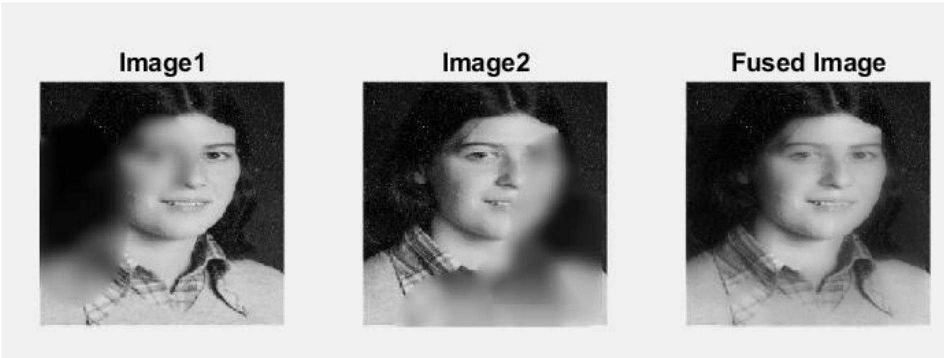
6. Results/ Software Simulation and Discussion

The results of the fusion were evaluated by comparing the fused images against the original images. Figures showing the original images and the fused image were generated. The fused image showed improved details, with clearer edges and better contrast compared to the individual input images.

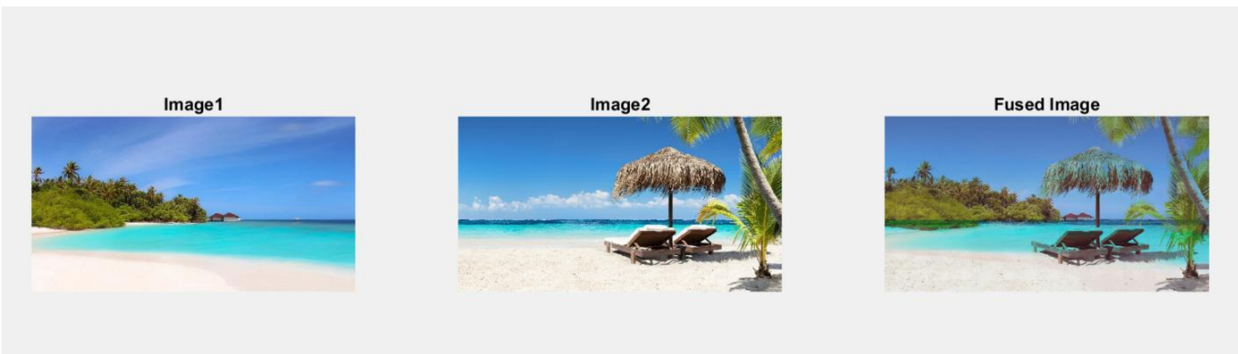
- Figure 1: Image 1
- Figure 2: Image 2
- Figure 3: Fused Image (using DWT fusion)

The test revealed that the MinMax, MinMin, MeanMax, MaxMax fusion strategy produced the most balanced results, maintaining both fine details and structural integrity in the fused image. The image fusion process performed efficiently even for large images, demonstrating the effectiveness of wavelet-based techniques in image processing.

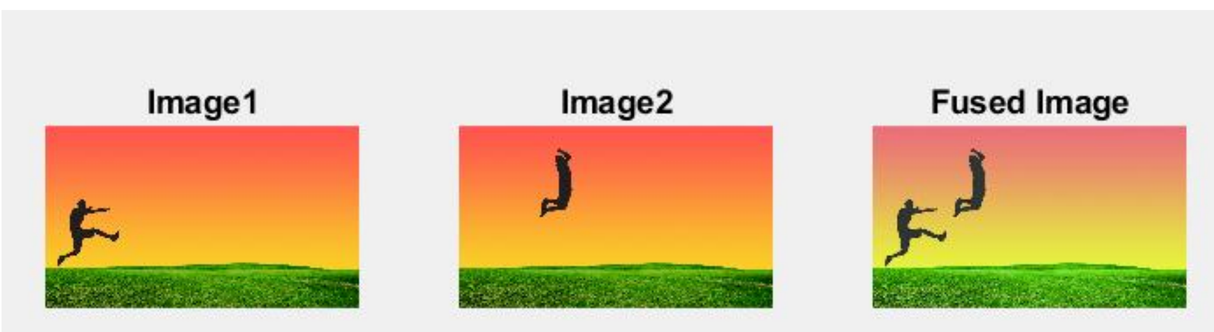
Output for MaxMax(code given above):



Output for MinMax(code provided on request):



Output for MinMin(code provided on request):



7. Conclusion:

In conclusion, this project successfully demonstrated the application of Discrete Wavelet Transform (DWT) for image fusion. The implemented system uses the multi-resolution properties of wavelets to fuse two images, preserving both high-frequency details (such as edges) and low-frequency components (such as structure). The fused image offers superior clarity and detail compared to the individual source images, making it useful for applications in fields like medical imaging, remote sensing, and computer vision. Further improvements could include experimenting with different wavelet functions and fusion strategies.

8. References:

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